MILITARY SPECIFICATION

INSIGNIA, EMBROIDERED (COAST GUARD)

This specification is approved for use by the Navy Clothing and Textile Research Facility, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for machine and hand embroidered insignia worn on the sleeve of enlisted personnel of the Coast Guard.

1.2 Classification. Embroidered insignia shall be of the following types and classes (see 6.2):

Type I - Rating badge
- Class 1 - Chief Petty Officer
- Class 2 - Petty Officer, First Class
- Class 3 - Petty Officer, Second Class
- Class 4 - Petty Officer, Third Class
- Class 5 - Senior Chief Petty Officer
- Class 6 - Master Chief Petty Officer
- Class 7 - Master Chief Petty Officer of the Coast Guard

Type II - Specialty Mark and Striker Mark

Type III - Group - Rate Mark
- Class 1 - One stripe
- Class 2 - Two stripes
- Class 3 - Three stripes

Type IV - Stripe, Length of Service

Type V - Branch of Service, Shield

Type VI - Insignia, Grade Star

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, and deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:53Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

V-T-276	- Thread, Cotton
PPP-B-676	- Boxes, Setup

MILITARY

MIL-C-823	- Cloth, Serge, Wool; Wool and Nylon; Polyester and Wool
MIL-L-17507	- Lace, Ornamental
MIL-E-17568	- Embroidery Materials, Metallic and Synthetic Metallic
MIL-B-17757	- Box, Shipping, Fiberboard (Modular Sizes)
MIL-C-29150	- Cloth, Knitted, Polyester, Twill, Double Knit
MIL-T-43548	- Thread, Polyester, Cotton-covered

STANDARDS

FEDERAL

FED-STD-191	- Textile Test Methods
FED-STD-751	- Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage

LAWS AND REGULATIONS

U. S. POSTAL SERVICE MANUAL

(Copies of the manual may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402.)

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

DOD STANDARD COLOR CARD OF OFFICIAL STANDARDIZED SHADES FOR HERALDIC YARNS FOR EMBROIDERY AND RIBBONS - 1974
The Color Association of the United States, Inc.

STANDARD SHADES FOR SEWING THREADS - 1968
The Color Association of the United States, Inc.

(Application for copies of the color cards should be addressed to the Color Association of the United States, Inc., 200 Madison Avenue, New York, NY 10016.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:53Z
Check the source to verify that this is the current version before use.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT
National Motor Freight Classification

(Application for copies shall be addressed to the American Trucking Association, Inc., Traffic Order Section, 1616 P Street, N.W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT
Uniform Freight Classification

(Application for copies shall be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, ILL 60606.)

3. REQUIREMENTS

3.1 Guide sample. Samples, when available, are solely for guidance and information to the contractor (see 6.3). Variation from this specification may appear in the sample, in which case this specification shall govern.

3.2 First article. When specified, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.2, and 6.2).

3.3 Materials.

3.3.1 Cloth, wool, serge. The serge cloth to be used as a background for embroidered designs shall conform to the requirements of type III, class 3 of MIL-C-823. The color shall match Blue 3362.

3.3.2 Cloth, double knit, polyester. The knit cloth to be used as a background for embroidered designs shall conform to the requirements of type I or II of MIL-C-29150. The color shall match Blue 3362.

3.3.3 Embroidery materials, metallic, and synthetic metallic. The embroidery materials shall conform to the requirements for type I, II, III, Classes 1 thru 6 of MIL-E-17568.

3.3.4 Lace, ornamental. The gold lace used for stripes and chevrons shall conform to the requirements for type II, style A, class 2, 3, 4, or 5, 3/8 inch width, one vellum of MIL-L-17507. The color shall match Gold 3903.

3.3.5 Buckram. The cotton buckram used as backing shall be natural or bleached white, plain weave, and shall conform to the following requirements when tested as specified in 4.3.1.

Type A - 50 (±3) ends and 50 (±3) picks per inch. The weight shall be 4.8 (±1.0) ounces per square yard.

Type B - 20 (±2) ends and 15 (±2) picks per inch. The weight shall be 6.75 (±1.0) ounces per square yard.

3.3.5.1 Buckram backing. Unless otherwise specified, buckram backing shall be used to reinforce the following areas of insignia:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:53Z
Check the source to verify that this is the current version before use.

Type A - Hand embroidered bullion or synthetic metallic stars, eagle, BUCKRAM specialty mark, and shield of rating badge. Sewn on gold lace chevrons and service stripes.

Type B - Two layers for Schiffli embroidered insignia on double knit BUCKRAM background fabrics.

3.3.6 Schiffli - Embroidery yarns.

3.3.6.1 Rayon yarn. The rayon face yarn shall be bright, continuous filament, regenerated cellulose type, 75/3 (± 5) denier Schiffli yarn. Testing shall be as specified in 4.3.1.

3.3.6.1.1 Colors. The colors of the rayon embroidery yarn shall be as shown below. The use of sulphur dyes is prohibited. The dyed yarn shall show good colorfastness to wet-dry cleaning, bleaching, dry and wet heat (hot pressing), light, and laundering when the end item is tested as specified in 4.3.4. Cable number for the yarn shall be as follows (see 2.2 and 6.2):

White	- Cable No. 67101
Scarlet	- Cable No. 67111
Emerald Green	- Cable No. 67128

3.3.6.1.2 Matching. The color shall match the approved standard shade sample under artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard shade sample under incandescent lamplight at 2800° Kelvin.

3.3.6.1.3 Bobbin thread. The bobbin thread used in Schiffli embroidery shall either be a good commercial grade of two-ply (minimum) undyed cotton thread or a continuous filament undyed (white) nylon thread equal to or better than the quality of the specified cotton thread.

3.3.7 Thread, cotton. The white or colored thread used for general stitching operations shall conform to type I requirements of V-T-276 and as specified in Table I. In addition, the colored threads shall show good colorfastness to dry and wet heat (hot pressing), when tested as specified in 4.3.1. Unless otherwise specified, the color of the thread shall match the color of the background cloth. As an alternate, the gold thread may be cotton-covered polyester conforming to ticket number 50, 2 or 3 ply of MIL-T-43548.

Table I - Threads

Type	Ply	Ticket No.		Operations
		Needle	Lower	
IA1	3	40	-	Attaching metallic embroidery materials by Hand
IC1	3	00	00	Hemming and cover stitching (white)
IC2	3	00	00	Hemming, attaching gold lace, coverstitching (colors)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:53Z
Check the source to verify that this is the current version before use.

3.3.8 Form, fiberboard. When needed, the form used to raise the design of hand embroidered metallic insignia shall be a commercial grade fiberboard of suitable thickness and shall be free of sulphur. The face side of the board shall be of a color to approximately match the color of the embroidery material covering it.

3.3.9 Stiffener. When needed, hand embroidered insignia may be treated on the reverse side with a stiffener to facilitate processing or to prevent raveling of the yarn. The stiffener shall not affect the color of the face of the insignia or the background fabric while in storage.

3.3.10 Stitch types and stitches per inch. Stitch types shall conform to FED-STD-751. Stitch type 301 used for hemming shall be 1/8 (± 1/32) inch from the edge. Stitch type 304 used for cover stitching shall be 1/8 (± 1/32) inch wide. Stitches per inch shall be 12 to 14 for general stitching and 45 to 48 for cover stitching.

3.4 Description. The insignia specified in this specification shall be machine embroidered on Schiffli type embroidery machines with rayon yarn in accordance with the applicable Government stitch enlargements (see 3.5.3) or hand embroidered with metallic (bullion) or synthetic metallic embroidery materials in accordance with approved guide samples (see 3.1).

3.4.1 Schiffli construction.

3.4.1.1 Ground and overlay stitches. The ground and overlay stitches shall be executed in accordance with the applicable stitch enlargement (see 3.5.3) and shall completely cover the prescribed background design area in a closely, densely, and tightly drawn manner to produce a clear, sharp execution of the design with none of the background cloth or underlay stitches visible.

3.4.1.1.1 Chevron (rating badge) and marks (group rate). Stitch execution of chevrons and marks shall be of the ground embroidery type. The number of visible rows shall be as indicated in the applicable stitch enlargement (see 3.5.3). The edges of each chevron and mark shall be finished with a 1/16 inch wide Stiele border stitch (see 3.4.1.3).

3.4.1.2 Underlay stitches. Underlay stitches, were required, shall be inserted in the direction and manner most suitable for proper design execution and shall not be less than the number indicated on the applicable stitch enlargement. The raised areas of the design where underlay is required, shall be equal to the guide sample (see 3.1) in fullness and depth.

3.4.1.3 Stiele border stitches. When required, Stiele border stitches shall be executed in accordance with the applicable stitch enlargement or guide sample (see 3.1 and 3.5.3) and shall be tightly drawn and well executed to produce a clear, sharp border and shall completely cover the background embroidery and cloth.

3.4.2 Hand construction with metallic (bullion) and/or synthetic metallic embroidery materials. When the eagle, stars, specialty marks, and shield are constructed by hand with metallic (bullion) and/or synthetic metallic embroidery materials the designs shall be produced using component materials specified in 3.3.3, 3.3.5.1, 3.3.8, 3.3.9 and Table II

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:53Z
Check the source to verify that this is the current version before use.

3.4.3 Gold lace chevrons and service stripes. Insignia utilizing gold lace shall be produced by combining the background cloth with type A buckram (see 3.3.5). The gold lace shall be superimposed over the background both in the prescribed manner (see 3.1) and stitched along the sides with thread of color to match the gold lace and using stitch type 301. The ends of the gold lace shall fit into slits cut into the background cloth and the entire length of slits completely coverstitched with stitch type 304 and thread matching the color of background cloth.

3.4.4 Finishing. Unless otherwise specified, all insignia shall have all outer edges of background cloth hemmed with a 3/8 to 1/2 inch turnunder, using stitch type 301 and seam Type EFa-1. The color of thread shall match the background cloth.

3.4.5 Spacing. All insignia covered by this specification shall be centered on the background fabric.

3.5 Color combination and design.

3.5.1 Color combination. The color combination of insignia shall be in accordance with Table II as indicated by the color combination code letter.

3.5.2 Design.

3.5.2.1 Schiffli embroidered. Stitch enlargements shall be furnished by the Government to the contractor for use in preparing the necessary punchings. The stitch enlargements shall not be altered in any way. All punchings shall be approved by the Government for conformity to requirements for minimum number of stitches as specified for the applicable insignia in Table III prior to start of production.

3.5.2.2 Hand embroidered. The design of the hand metallic or synthetic metallic embroidered and/or gold lace insignia shall be in accordance with the Government approved guide sample and shall not be altered in any way (see Table IV).

3.5.2.3 Direction of design. Background fabrics having a twill effect in their weave shall have the height dimension of the embroidered design finishing in the length (warp or wale) direction of the fabric.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:53Z
Check the source to verify that this is the current version before use.

TABLE II - Color and Material for Design

Color Combination Code Letter	Background Cloth	Rating Badge				Type of Design			
		Eagle	Specialty Mark	Chevrons	Stars	Striker Mark Specialty	Group Rate	Service Stripe	Shield
CG-06 (E10)	Blue Serge	Silver Bullion or Syn Met	Gold Bullion or Syn Met	Gold Syn Lace	Gold Bullion or Syn Met	N/A	N/A	Gold Syn. Lace	Gold Bullion or Syn Met
CG-01 (E7-E9)	Blue Serge	Silver Bullion or Syn Met	Silver Bullion or Syn Met	Gold Syn. Lace	Silver Bullion or Syn Met	N/A	N/A	Gold Syn. Lace	Gold Bullion or Syn Met
CG-02 (E4-E6)	Blue Serge	White	White	Scarlet	N/A	N/A	N/A	Scarlet	Gold Bullion or Syn Met
CG-03	Blue Serge	N/A	White	N/A	N/A	White	White	N/A	Gold Bullion or Syn Met
CG-04	Blue Serge	N/A	White	N/A	N/A	White	Scarlet	N/A	Gold Bullion or Syn Met
CG-05	Blue Serge	N/A	White	N/A	N/A	White	Emerald Green	N/A	Gold Bullion or Syn Met

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:53Z
Check the source to verify that this is the current version before use.

TABLE II - Color and Material for Design (cont'd)

Color Combination Code Letter	Background Cloth	Rating Badge				Type of Design			
		Eagle	Specialty Mark	Chevrons	Stars	Striker Mark Specialty	Group Rate	Service Stripe	Shield
Z10 (CG) (E10)	Blue Double Knit	Silver Bullion or Syn Met	Gold Bullion or Syn Met	Gold Syn Lace	Gold Bullion or Syn Met	N/A	N/A	Gold Syn Lace	Gold Bullion or Syn Lace
AAA (CG) (E7-E9)	Blue Double Knit	Silver Bullion or Syn Met	Silver Bullion or Syn Met	Gold Syn Lace	Silver Bullion or Syn Met	N/A	N/A	Gold Syn Lace	Gold Bullion or Syn Met
WWW (CG) (E4-E6)	Blue Double Knit	White	White	Scarlet	N/A	N/A	N/A	Scarlet	Gold Bullion or Syn Met
Z3 (CG)	Blue Double Knit	N/A	White	N/A	N/A	White	White	N/A	Gold Bullion or Syn Met
Z4 (CG)	Blue Double Knit	N/A	White	N/A	N/A	White	Scarlet	N/A	Gold Bullion or Syn Met
Z5 (CG)	Blue Double Knit	N/A	White	N/A	N/A	White	Emerald Green	N/A	Gold Bullion or Syn Met

NOTES: (a) Abbreviation N/A = Not Applicable
(b) Wherever white, scarlet, and emerald green embroidery material is indicated, rayon yarn is required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:53Z
Check the source to verify that this is the current version before use.

3.5.3 Construction requirements. Construction requirements and dimensions for Schiffli and hand embroidered insignia shall be as specified in Tables III and IV.

TABLE III - Requirements for Schiffle embroidered insignia

Type and class	Minimum No. of stitches	2/ Design and dimensions	Finished sizes (inches)	Stitch enlargement drawing No.
Type I, Class 2, Petty Officer, First Class	6980 1/	see figure 5	4x 5 7/8	16-1-372
Type I, Class 3, Petty Officer, Second Class	6268 1/	see figure 6	4x 5 1/8	16-1-372
Type I, Class 4, Petty Officer, Third Class	5556 1/	see figure 7	4x 4 3/8	16-1-372
Type II, Specialty Mark or Striker Mark				
Aviation Electrician's Mate	895	see figure 8	2 7/8x 1 7/8	16-1-286
Aviation Electronic's Technician	825	see figure 8	2 7/8x 1 7/8	16-1-208
Aviation Machinist's Mate	960	see figure 8	2 7/8x 1 7/8	16-1-287
Aviation Structural Mechanic	910	see figure 8	2 7/8x 1 7/8	16-1-291
Aviation Survivalman	1439	see figure 8	2 7/8x 1 7/8	16-1-135
Boatswains Mate	580	see figure 8	2 x 2	16-1-293
Damage Controlman	405	see figure 8	2 x 2	16-1-442
Data Processing Techician	550	see figure 8	2 x 2	16-1-24
Dental Technician	755	see figure 8	2 x 2	16-1-301
Electrician's Mate	1280	see figure 8	2 x 2	16-1-204
Electronic's Technician	450	see figure 9	2 x 2	16-1-221
Machinery Technician	690	see figure 9	2 x 2	16-1-303
Fire Control Technician	540	see figure 9	2 x 2	16-1-306
Fire Fighter	1243	see figure 9	2 x 2	16-1-590
Gunner's Mate	515	see figure 9	2 x 2	16-1-307
Hospital Corpsman	685	see figure 9	2 x 2	16-1-203
Marine Science Technician	615	see figure 9	2 x 2	16-1-134
Musician	510	see figure 9	2 x 2	16-1-317
Photo Journalist	594	see figure 9	2 x 2	16-1-443
Port Securityman	775	see figure 9	2 x 2	16-1-589
Quartermaster	590	see figure 10	2 x 2	16-1-323
Radarman	410	see figure 10	2 x 2	16-1-324
Radioman	625	see figure 10	2 x 2	16-1-325
Sonar Technician	525	see figure 10	2 7/8x 1 7/8	16-1-528
Storekeeper	550	see figure 10	2 x 2	16-1-332
Subsistence Specialist	641	see figure 10	2 x 2	16-1-444
Telephone Technician	1160	see figure 10	2 x 2	16-1-308
Yeoman	300	see figure 10	2 x 2	16-1-210

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:53Z
Check the source to verify that this is the current version before use.

TABLE III - Requirements for Schiffli embroidered insignia (cont'd)

Type and class	Minimum No. of stitches	2/ Design and dimensions	Finished sizes (inches)	Stitch enlargement drawings No.
Type III, Group-Rate Mark				
Class 1-One stripe	890	see figure 11	3 1/8 x 2 5/16	16-1-15
Class 2-Two stripes	1780	see figure 11	3 1/8 x 2 15/16	16-1-15
Class 3-Three stripes	2670	see figure 11	3 1/8 x 3 1/2	16-1-15

1/ Exclusive of the number of stitches for the speciality mark.
2/ Figures 5 thru 11 are for dimensions only and are not drawn to scale. For correct design and proper stitch execution refer to the applicable stitch enlargement. Unless otherwise specified, all dimensions are in inches.

TABLE IV - Requirements for hand embroidered insignia

Type and class	1/ Design and dimensions	Finished sizes (inches)
Type I		
Class 1, Chief Petty Officer	see figure 4	4 x 5 7/8
Class 5, Senior Chief Petty Officer	see figure 3	4 x 7 1/4
Class 6, Master Chief Petty Officer	see figure 2	4 x 7 1/4
Class 7, Master Chief Petty Officer of the CG	see figure 1	4 x 7 1/4
Type IV Stripe, Length of Service (Gold Lace)		
One stripe	see figure 12	6 x 7/8
Two stripes	see figure 12	6 x 1 1/2
Three stripes	see figure 12	6 x 2 1/8
Four stripes	see figure 12	6 x 2 3/4
Five stripes	see figure 12	6 x 3 3/8
Six stripes	see figure 12	6 x 4
Seven stripes	see figure 12	6 x 4 5/8
Type V, Shield, Branch of Service	see figure 13	1 x 13/16
Type IV, Insignia, Grade, Star	see figure 13	3/4 diameter

1/ Figures 1, 2, 3, 4, 12, and 13 are for dimensions only and are not drawn to scale.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:53Z
Check the source to verify that this is the current version before use.

3.6 Type I, Rating badge.

3.6.1 Construction (machine embroidered Schiffli). Type I, Classes 2, 3, and 4 rating badges shall be embroidered on the background cloth specified in table II in accordance with the applicable stitch enlargement listed in table III, the dimensions specified in the applicable figure, and the construction requirements of 3.4.1.

3.6.2 Construction (Hand embroidered). Type I, classes 1, 5, 6, and 7 rating badges shall be embroidered on the background cloth specified in Table II in accordance with the approved guide sample, the dimensions specified in the applicable figure, and the construction requirements of 3.4.2 and table V.

3.6.3 Type II - Specialty Mark and Striker Mark. The specialty mark shall be centered on the rating badge between the upper chevron and the eagle's talons. Striker mark shall be centered on the background cloth specified in Table II and shall be embroidered in accordance with the designs and dimensions indicated for specialty marks and striker marks in figures 8 thru 10. Construction shall be in accordance with the applicable requirements of 3.6.1 or 3.6.2.

3.6.4 Type III - Group Rate Mark. The group rate marks shall consist of one (class 1) stripe two (class 2) stripes or three (class 3) stripes embroidered diagonally on the background cloth specified in Table II. The stripes shall be machine embroidered in accordance with the applicable stitch enlargement listed in Table III, the dimensions specified in figure 11, and the construction requirements of 3.4.1.

3.6.5 Type IV - Stripe, Length of Service. Insignia shall consist of one thru seven stripes, as applicable, stitched diagonally on the background cloth specified in Table II. The gold lace stripes shall be cut to meet the dimensions specified in figure 12, and the construction requirements of 3.4.3.

3.6.6 Type V - Insignia, Branch of Service, Shield. The shield shall be hand embroidered with 13 simulated stars at top one-third of shield, arranged in two horizontal rows, 7 stars over 6 stars. The bottom two-thirds of shield shall have 12 or 13 vertical stripes. Shield shall be hand embroidered on the background cloth specified in Table II and in accordance with the approved guide sample, the dimensions specified in figure 13, and the construction requirements of 3.4.2 and 3.6.2.

3.6.7 Type VI - Insignia, grade, star. The star(s) shall be of a 5-point design in accordance with the dimensions specified in figure 13. Star(s) shall be hand embroidered in accordance with 3.4.2 and 3.6.2.

3.7 Measurement Tolerances. The tolerances listed in Table V shall be applicable to specified dimensions for finished insignia.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:53Z
Check the source to verify that this is the current version before use.

Table V - Tolerances

Inches		Tolerances
Dimensions	under 15/16	± 1/32
Dimensions ranging -	from 1 to 1 15/16	± 1/16
	from 2 to 2 15/16	± 3/32
	from 3 to 3 15/16	± 1/8
	from 4 to 4 15/16	± 5/32
	from 5 to 5 15/16	± 3/16
	from 6 to 6 15/16	± 7/32
	from 7 to 7 15/16	± 1/4
Degrees for angle of chevrons		± 3

3.8 Workmanship. The finished insignia shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificates of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 First article inspection. When required (see 6.2) the first article submitted in accordance with 3.2 shall be inspected as specified in 4.3.2 for compliance with design, construction, workmanship, and dimensional requirements.

4.3 Inspection. Inspection shall be in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated herein.

4.3.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document. In addition, components and materials listed in Table VI shall be tested for the characteristics shown by following test methods specified in FED-STD-191, wherever applicable. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if the average of one or more sample units fail to meet the requirement specified. The sample unit and sample size for testing components shall be as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:53Z
Check the source to verify that this is the current version before use.

Component	Lot Size Expressed as	Sample Unit for Component Testing
Buckram	Yards	1 Linear Yard
Embroidery yarn	Holders	1 cone or spool, or sufficient number of bobbins to total a minimum of 360 yards
Thread (all types and sizes	Holders	1 cone or spool, or sufficient number of bobbins to total a minimum of 360 yards

Sample Size for Component Testing

Lot size	Sample Size (units)
800 or less	2
801 thru 22,000	3
22,001 and over	5

Table VI - Component Testing

Component	Characteristic	Requirement paragraph	Test method	Determinations per unit of product	Results reported as
Buckram (Types A and B	Material identification	3.3.5	1/	1	Pass or fail
	Yarn per inch				
	Ends	3.3.5	5050	5	Pass or fail
	Picks	3.3.5	5050	5	Pass or fail
	Weight	3.3.5	5041	5	Pass or fail
	Weave	3.3.5	Visual	1	Pass or fail
Rayon Embroidery yarn	Material identification	3.3.6.1	1/	1	Pass or fail
	Denier	3.3.6.1	4021	1	Pass or fail
	Ply	3.3.6.1	Visual	1	Pass or fail
	Presence of sulfur	3.3.6.1.1	2020 1/	1	Pass or fail
Bobbin thread	Material identification	3.3.6.1.3	1/	1	Pass or fail
	Ply	3.3.6.1.3	Visual	1	Pass or fail
Thread, cotton	Colorfastness to:				
	Dry heat (hot pressing)	3.3.7	5640	1	Pass or fail
	Wet heat (hot pressing)	3.3.7	5640	1	Pass or fail
Fiberboard Form	Presence of sulfur	3.3.8	2020 1/	1	Pass or fail

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:53Z
Check the source to verify that this is the current version before use.

4.3.2 Examination of the end item. The defects found during the examination of the end item shall be classified in accordance with 4.3.2.1 and 4.3.2.2. The sample unit shall be one finished insignia. The lot size shall be expressed in units of one insignia. The inspection levels and acceptable quality levels shall be as specified in 4.3.2.3.

4.3.2.1 Visual examination. The finished insignia shall be examined for defects in design, material, construction and workmanship and classified in accordance with the defects listed in Table VII.

Table VII - Classification of defects

Examine		Defects
I Materials (Gold Lace and background cloth)	(a)	Other than specified.
	(b)	Color or shade other than specified.
	(c)	Hole, float, slub, knot, broken yarn, spot or stain
II Design	(a)	Any deviation from applicable stitch enlargement, drawing, illustration or guide sample.
	(b)	Wrong color or wrong color arrangement.
	(c)	Not centered in the prescribed background.
	(d)	Form board used to raise design missing (when applicable).
III Construction	(a)	Stitching incomplete
	(b)	Embroidered on wrong side of background cloth.
	(c)	Broken or skipped stitches, any
	(d)	Shifting of yarns, affecting appearance
	(e)	Incomplete coverage, background cloth or underlay stitches visible.
	(f)	Density not in accordance with guide sample.
	(g)	Bobbin threads visible, affecting appearance.
	(h)	Bunched or tangled threads and/or yarns.
	(i)	Loose or tight tension, affecting appearance.
	(j)	Puckering, affecting appearance
	(k)	Stitches out of alignment, affecting appearance.
	(l)	Open or poorly formed points in gold lace.
	(m)	Uneven spacing between gold lace service stripes or chevrons
	(n)	Buckram omitted, incomplete, or extending beyond insignia
	(o)	Height dimension of embroidered design not finishing in length direction of (when applicable) background cloth having a twill effect in its weave.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:53Z
Check the source to verify that this is the current version before use.

Table VII - Classification of defects (Cont.)

Examine		Defects
IV Finish Stitching and Attachment	(a)	Stitches per inch not within specified range
	(b)	Tight or loose tension affecting appearance.
	(c)	Irregular or unevenly gauged stitching, affecting appearance
	(d)	Broken or skipped stitches extending more than 1/4 inch along stitchline
	(e)	Color of hemming thread not as specified
	(f)	Ends not fitted into slits cut in background cloth (i.e., gold lace chevrons and service stripes).
	(g)	Points not mitered (i.e., gold lace chevrons)
	(h)	Edges not finished as required.
	(i)	Wrong stitch type.
	(j)	Wrong seam type.
	(k)	Stitching run-off
V Workmanship	(a)	Spot or stain
	(b)	Misshaped or distorted.
	(c)	Two or more thread ends longer than 1/16 inch
	(d)	Color of insignia face affected by stiffener coating.

4.3.2.2 Dimensional examination. The finished insignia shall be examined for defects in dimension. Any measurement deviating from specified dimensions or tolerances specified on applicable figure and in Table V.

4.3.2.3 Inspection levels and acceptable quality levels. The inspection levels and acceptable quality levels (AQL's) expressed as defects per 100 units for visual and dimensional examination shall be as follows:

	Inspection level	AQL
For visual examination in 4.3.2.1	II	4.0
For dimensional examination in 4.3.2.2	S-3	6.5

4.3.3 Examination of packaging. Examination shall be made to determine that packaging, packing and marking comply with the Section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packed, selected just prior to the closing operation. Defects of closure listed below shall be examined on shipping containers fully packaged. The lot size shall be the number of containers in the inspection lot. The sample size shall be in accordance with level S-2 of MIL-STD-105. The acceptable quality level (AQL) for this examination shall be 2.5 defects per hundred units.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:53Z
Check the source to verify that this is the current version before use.

Examine	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence or method of application.
Materials	Any component missing, or not as specified. Any component damaged, affecting serviceability.
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, loose strapping inadequate stapling. Bulging or distortion of container.
Content	Number of items per container is more or less than specified. Wrong type, class or styles per set-up box. Number of set-up boxes per shipping container more or less than specified.

4.3.4 Testing of the end item. The end item shall be tested as specified in Table VIII. The lot size shall be expressed in units of one insignia. The unit of product for test purposes shall be six insignia. The lot shall be unacceptable if one or more units fail to meet the requirements referenced in Table VIII. The sample size shall be as follows:

Lot size (each)	Sample size
800 or less	2
801 thru 22,000	3
22,001 and over	5

TABLE VIII - End item testing

Characteristic	Requirement paragraph	Test method	No. of determinations per unit of product	Results reported as
Rayon yarn Colorfastness to: 1/ 2/		3/		
Bleaching	3.3.6.1.1	5600	1	Pass or fail
Dry heat (hot pressing)	3.3.6.1.1	5640	1	Pass or fail
Wet heat (hot pressing)	3.3.6.1.1	5640	1	Pass or fail
Light	3.3.6.1.1	5660	1	Pass or fail
Laundering	3.3.6.1.1	5614	1	Pass or fail
Wet-dry cleaning	3.3.6.1.1	5622	1	Pass or fail
Shade match of Rayon Yarn	3.3.6.1.2	Visual	Each Color	Pass or fail
Type of base cloth	3.3.1 and 3.3.2	Visual	1	Pass or fail

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:53Z
Check the source to verify that this is the current version before use.

1/ Test reports will be furnished for each color of yarn used in the initial shipping lot of finished insigina. A contractor's certificate of compliance will be accepted for each color for each subsequent lot of insignia on the same contract, provided the contractor certifies that the dyer or dye formulation has not been changed.

2/ When base cloth specification does not specify a colorfastness test for laundering, table VIII requirements for fastness to bleaching and laundering shall not apply.

3/ Except that the test specimens shall be the finished embroidered insignia.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Quantity per set-up box.

5.1.1.1.1 Type I - Badge, Rating - All Classes. Each rating badge shall be individually packaged in a snug fitting, flat style polyethylene bag that conforms to the requirements specified in paragraph 5.1.1.2. The bag opening shall be across the width of the bag face. The open end of the bag shall be folded over. Twenty (20) rating badges of one class only shall be packaged in a set-up box that conforms to the requirements specified in paragraph 5.1.1.3.1.

5.1.1.1.2 Type II - Specialty Mark and Striker Mark. Ten (10) marks, of one kind only, shall be packaged in a snug fitting, flat style polyethylene bag that conforms to the requirements specified in paragraph 5.1.1.2. The bag opening shall be across the width of the bag face. The open end of the bag shall be folded over. One hundred twenty (120) marks (12 bags of 10 each), of one kind only shall be packaged in a set-up box that conforms to the requirements specified in paragraph 5.1.1.3.1.

5.1.1.1.3 Type III - Group Rate Mark - All Classes. Ten (10) marks, of one kind only, shall be packaged in a snug fitting flat style polyethylene bag that conforms to the requirements specified in paragraph 5.1.1.2. The bag opening shall be across the width of the bag face. The open end of the bag shall be folded over. Sixty (60) marks (6 bags of 10 each) shall be packaged in a set-up box that conforms to the requirements specified in paragraph 5.1.1.3.1.

5.1.1.1.4 Type IV - Stripe, Length of Service

240 each (1 stripe insignia)
150 each (2 stripe insignia)
90 each (3 stripe insignia)
60 each (4 or 5 stripe insignia)
30 each (6 or 7 stripe insignia)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:53Z
Check the source to verify that this is the current version before use.

The above quantities of service stripes, of one kind only, shall be packaged in a set-up box that conforms to the requirements specified in 5.1.1.3.2. Thirty (30) items of one kind only, shall be tied at each end by elastic band, twine or cloth tape to form a bundle.

5.1.1.1.5 Type V - Insignia, Branch of Service, Shield. Five (5) insignia shall be packaged in a snug fitting, flat style polyethylene bag that conforms to the requirements specified in 5.1.1.2. The bag opening shall be across the width of the bag face. The open end of the bag shall be folded over. Sixty (60) insignia (12 bags of 5 each), shall be packaged in a set-up box that conforms to the requirements specified in 5.1.1.3.1.

5.1.1.1.6 Type VI - Insignia, Grade, Star. The insignia shall be packaged in accordance with the industry's practice.

5.1.1.2 Bags, polyethylene. The polyethylene bags shall be fabricated from clear polyethylene film of 0.0010 inch thickness (+ 20% tolerance). The polyethylene bags shall be formed with heat-sealed seams that are straight, continuous, and parallel to each other and the formed edges of the bag.

5.1.1.3 Boxes, set-up.

5.1.1.3.1 Types I, II, III, V, and VI Insignia. Items, of one type and kind only, shall be placed in a two-piece, full telescope style, set-up paperboard box conforming to type I, variety 1, class A, style 4 of PPP-B-676. The outside dimensions of the set-up box shall be 7 inches in length, 5 inches in width, and 2 1/8 inches in depth. Unless otherwise specified (see 6.2), the interior of set-up boxes containing wool or part wool items shall be sprinkled with not less than 1/4 ounce of flaked naphthalene.

5.1.1.3.2 Type IV Insignia. Items, of one kind only, shall be placed in a two-piece, full telescope style, set-up paperboard box conforming to type I, variety 1, class A, style 4 of PPP-B-676. The outside dimensions of the set-up box shall be 11 1/2 inches in length, 7 5/8 inches in width and 2 3/8 inches in depth. Unless otherwise specified (see 6.2.1), the interior of set-up boxes containing wool or part wool items shall be sprinkled with not less than 1/4 ounce of flaked naphthalene.

5.1.2 Level C (Commercial packaging). Insignia shall be packaged to afford adequate protection against physical damage during shipment from the contractor to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the contractor for retail distribution. Containers shall comply with the U.S. Postal Service Manual, Uniform Freight Classification Rules or National Freight Classification Rules, as applicable.

5.2 Packing. Packing shall be level A, B or C (see 6.2).

5.2.1 Level A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:53Z
Check the source to verify that this is the current version before use.

5.2.1.1 Types I, II, III, V and VI Insignia. Fifteen (15) set-up boxes of one type, class or kind of insignia packaged as specified in 5.1.1, shall be packed in a shipping container conforming to size 1, class weather-resistant, variety SW or DW of MIL-B-17757. Each shipping container shall be closed, waterproofed and reinforced in accordance with the appendix of the box specification.

5.2.1.2 Type IV Insignia. Six (6) set-up boxes of one class and kind packaged as specified in 5.1.1, shall be packed in a shipping container conforming to size 1, class weather-resistant variety SW or DW of MIL-B-17757. Each shipping container shall be closed, waterproofed and reinforced in accordance with the appendix of the box specification.

5.2.2 Level B.

5.2.2.1 Types I, II, III, V and VI Insignia. Fifteen (15) set-up boxes of one type, class and kind of insignia, packaged as specified in 5.1.1, shall be packed in a shipping container conforming to size 1, class domestic, variety SW or DW, grade 200 of MIL-B-17757. Shipping containers shall be closed in accordance with the appendix of the box specification.

5.2.2.2 Type IV Insignia. Six (6) set-up boxes of one class and kind packaged as specified in 5.1.1, shall be packed in a shipping container conforming to size 1, class domestic, variety SW or DW, grade 200 of MIL-B-17757. Shipping containers shall be closed in accordance with the appendix of the box specification.

5.2.2.3 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container for the finished insignia shall be a grade V3c, W5c, or W6c fiberboard box fabricated in accordance with MIL-B-17757 and closed in accordance with the appendix of the box specification.

5.2.3 Level C (Commercial packing). Finished insignia, packaged as specified in 5.1, shall be packed in a manner to insure safe delivery at destination at the lowest transportation rate for such deliveries. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall comply with the U.S. Postal Service Manual, Uniform Freight Classification Rules, or National Motor Freight Classification Rules, as applicable.

5.3 Marking. In addition to any special marking required by the contract or order, intermediate packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The embroidered insignia covered by this specification are intended to designate the rate, specialty, grade, apprenticeship, and length of service of enlisted personnel in the Coast Guard.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:53Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Type, class, and color combination required (see 1.2 and 3.5).
c. When first article sample is required (see 3.2).
d. Cable number for yarn (see 3.3.6.1.1).
e. Levels of packaging and packing required (see 5.1 and 5.2).
f. When weather-resistant fiberboard shipping containers are required for level B packing (see 5.2.2.3).

6.2.1 The naphthalene flakes requirement is not applicable when Government furnished material of wool or part wool content has been mothproof treated (see 5.1.1.3.1 and 5.1.1.3.2).

6.3 Samples. For access to guide samples address the procuring office issuing the invitation for bids.

Custodian:
Navy - NU

Review activities:
Navy - CG
DLA - CT

Preparing activity:
Navy - NU

Project No. 8455-N948

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:53Z
Check the source to verify that this is the current version before use.

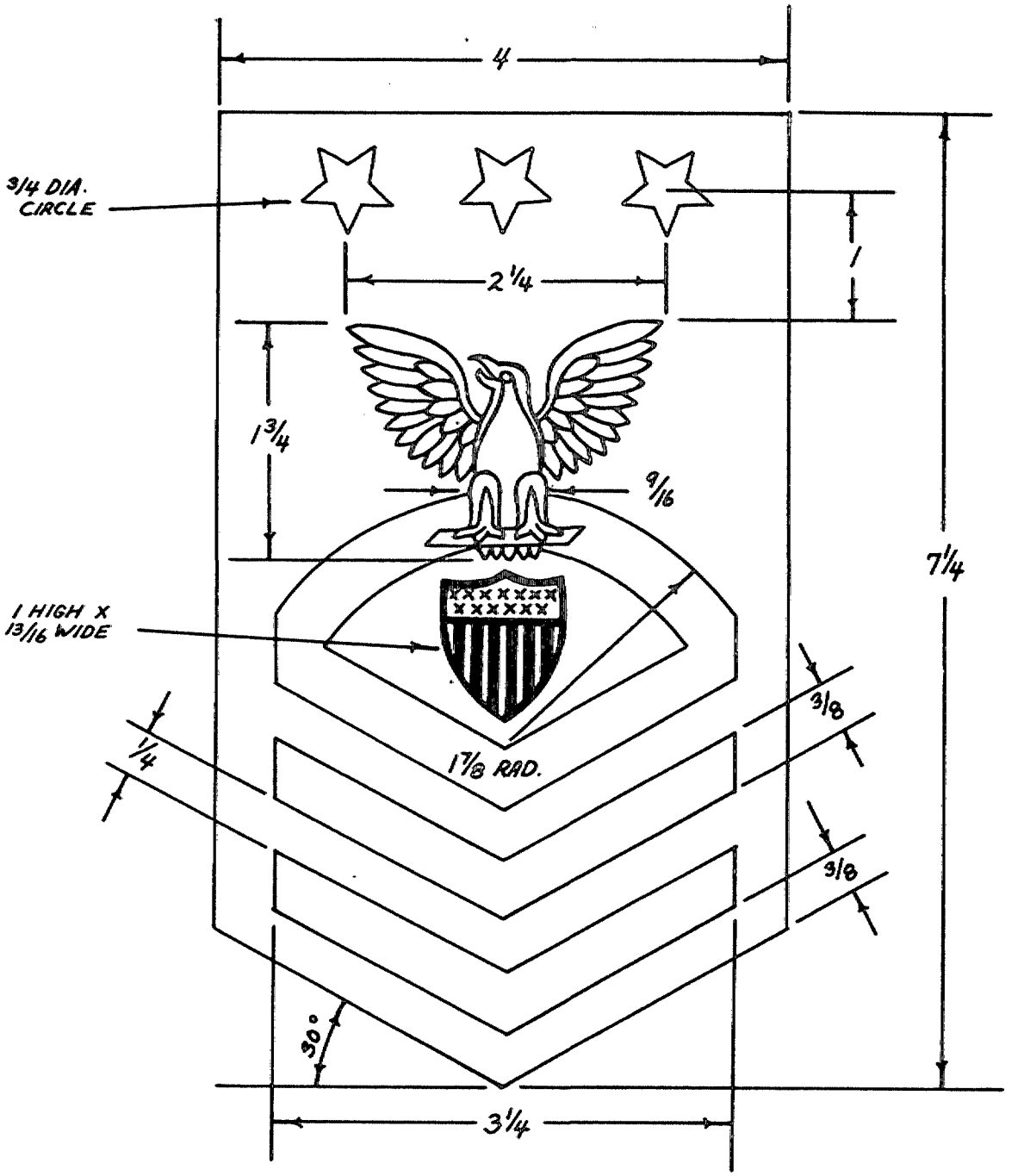


FIG. 1 - RATING BADGE, MASTER CHIEF PETTY OFFICER OF THE COAST GUARD

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:53Z
Check the source to verify that this is the current version before use.

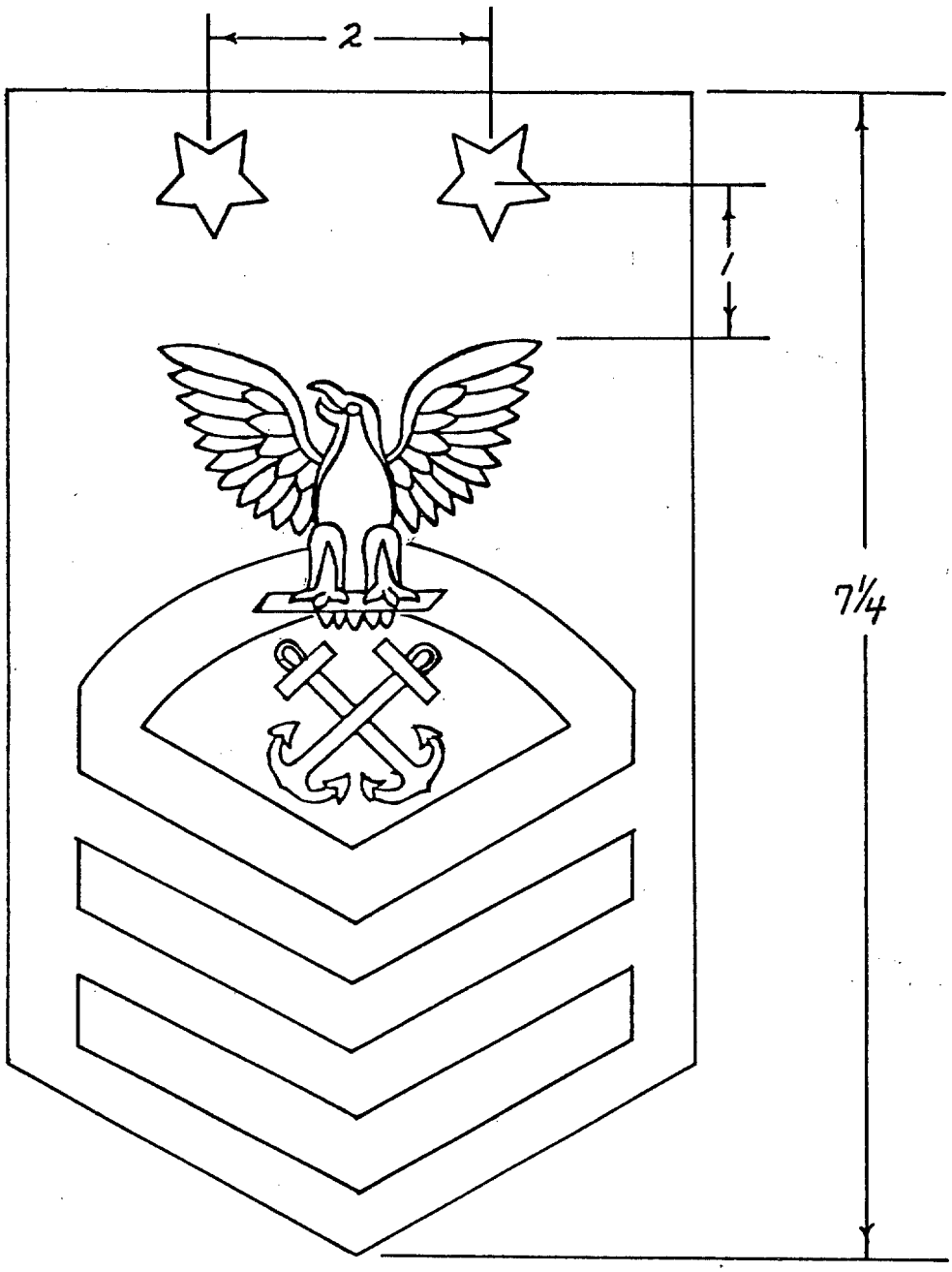


FIG. 2 -RATING BADGE, MASTER CHIEF PETTY OFFICER

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:53Z
Check the source to verify that this is the current version before use.

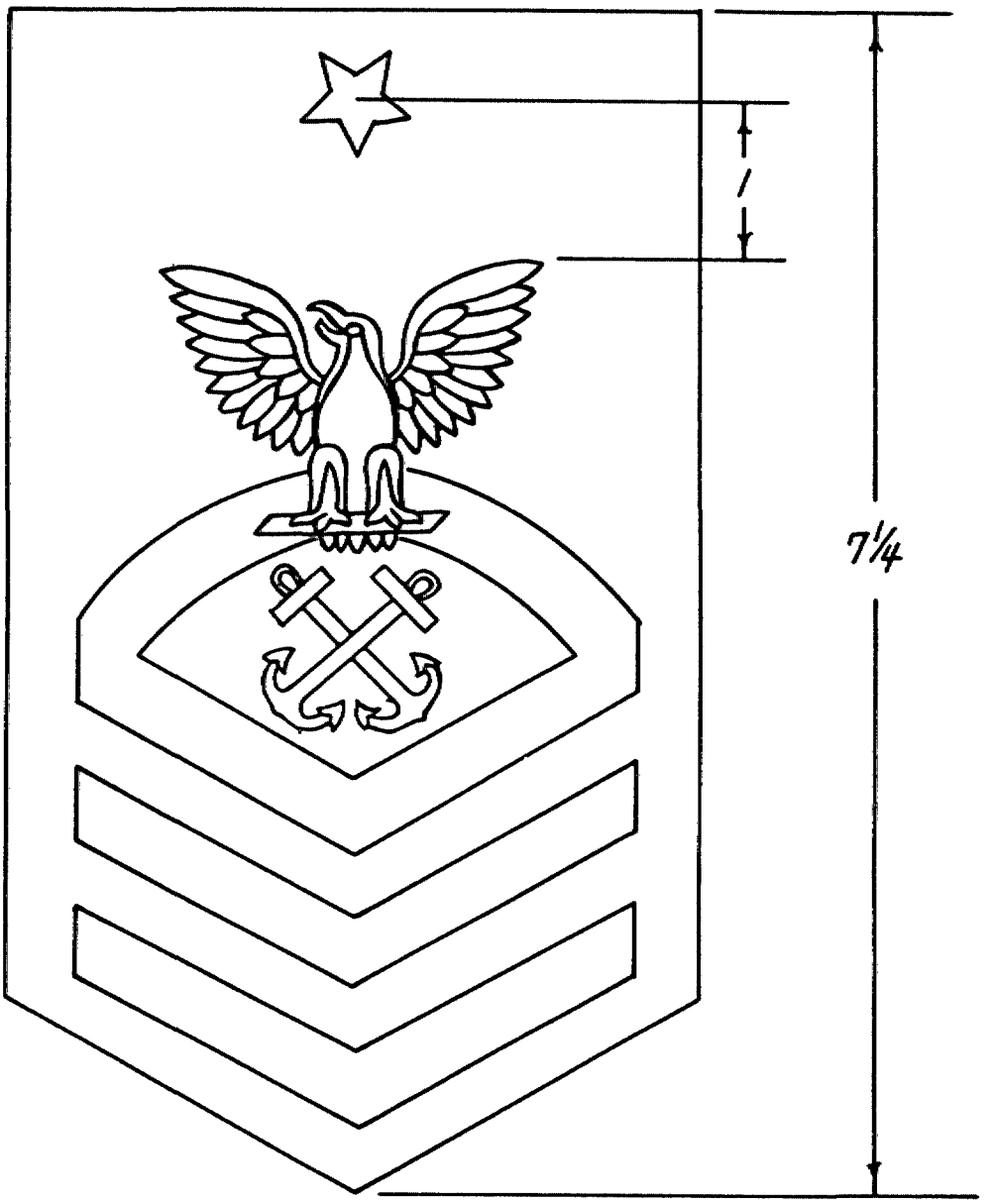


FIG. 3- RATING BADGE, SENIOR CHIEF PETTY OFFICER

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:53Z
Check the source to verify that this is the current version before use.

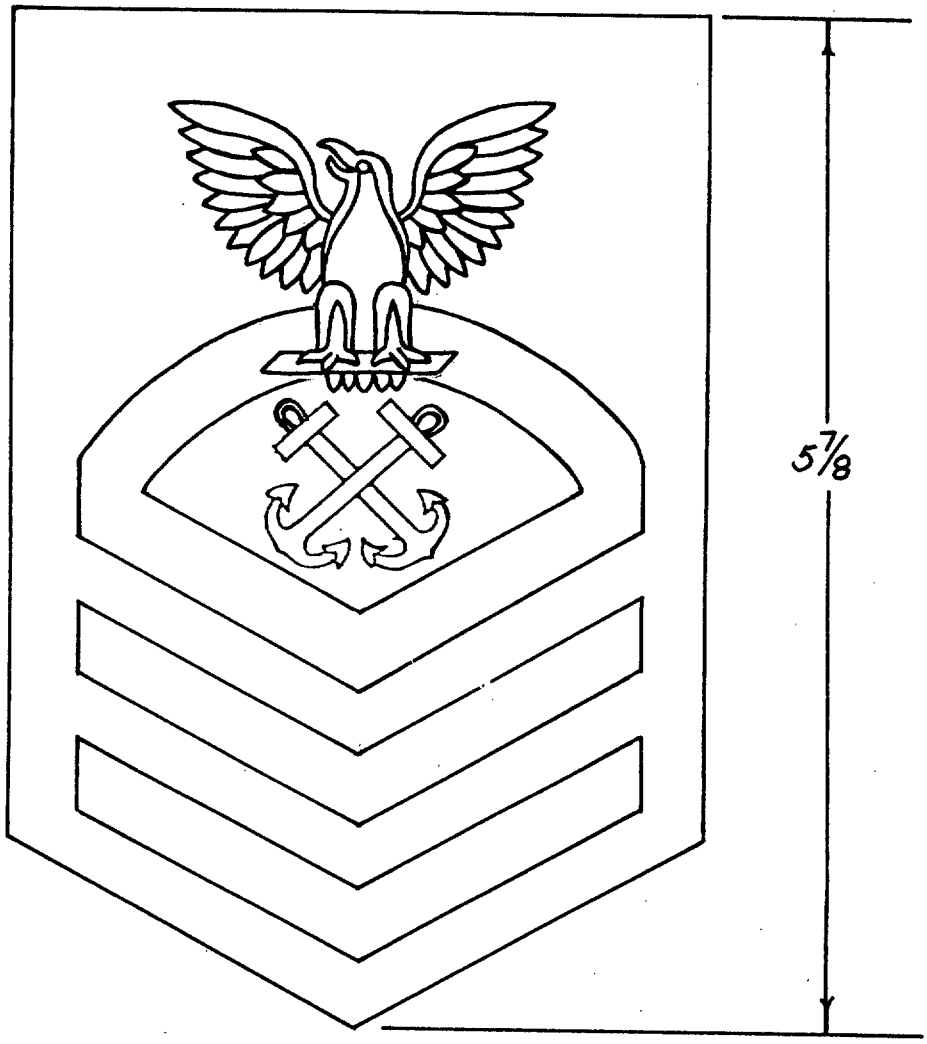


FIG. 4 - RATING BADGE, CHIEF PETTY OFFICER

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:53Z
Check the source to verify that this is the current version before use.

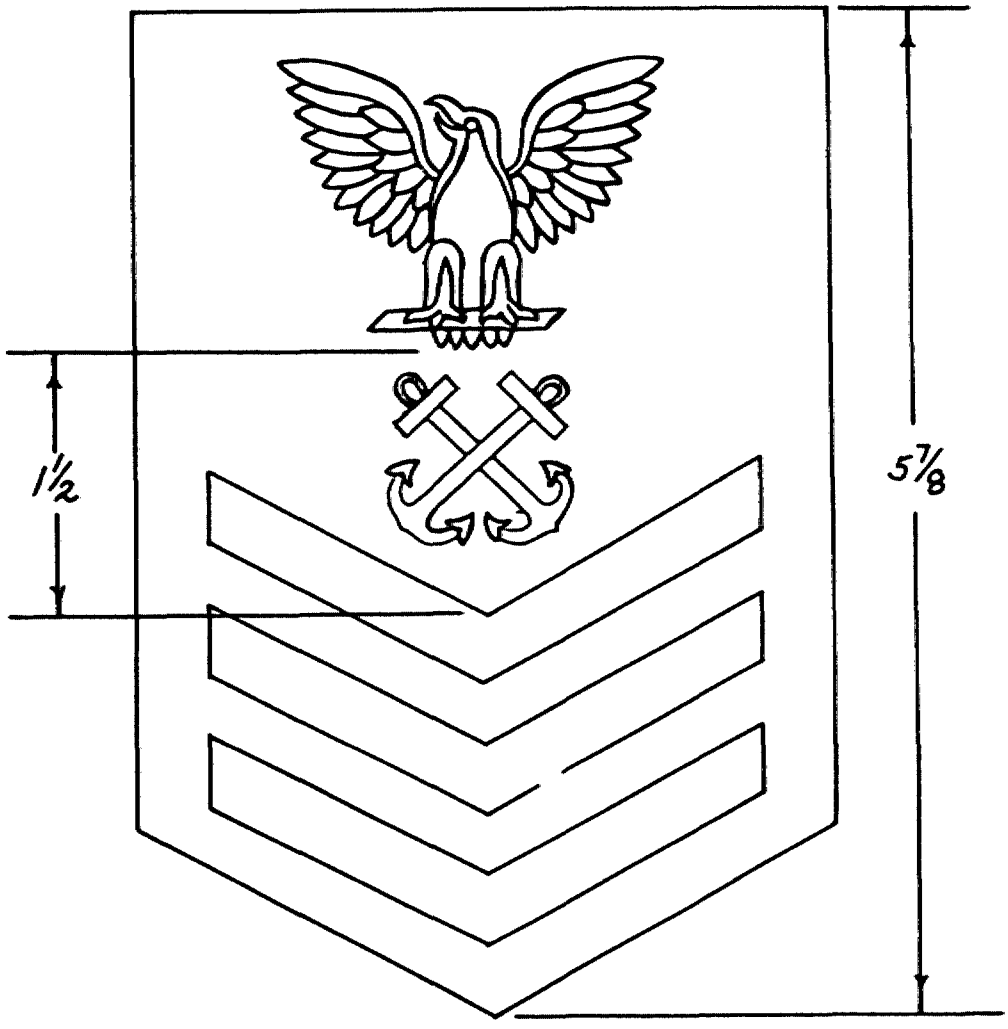


FIG. 5 - RATING BADGE, PETTY OFFICER, FIRST CLASS

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:53Z
Check the source to verify that this is the current version before use.

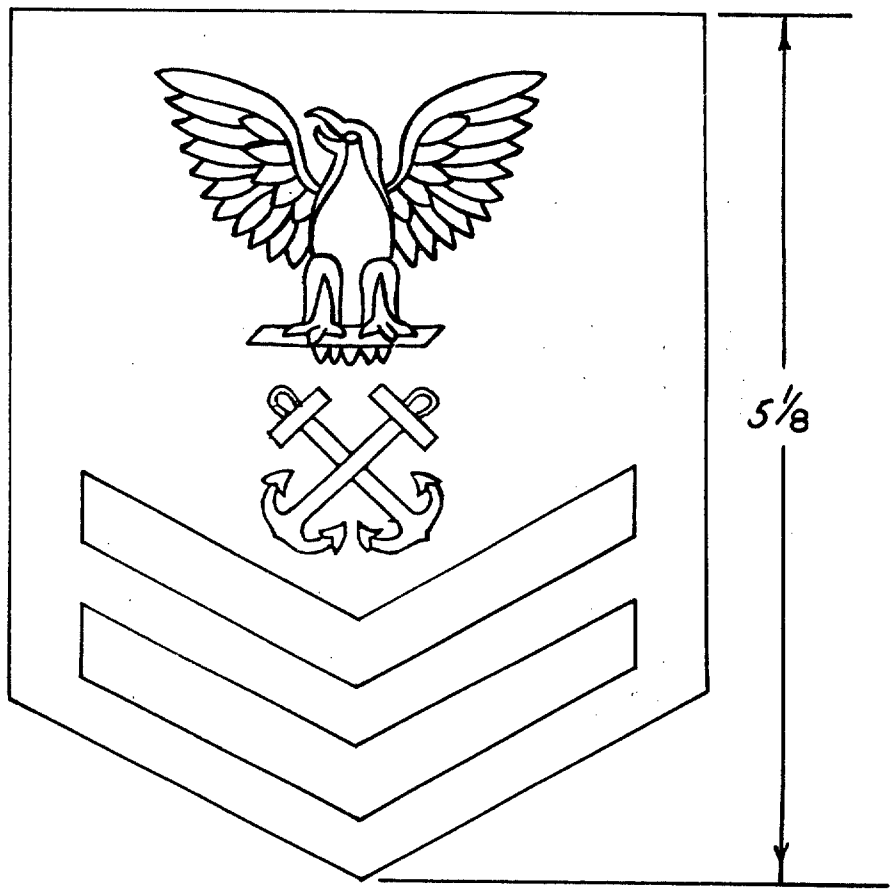


FIG. 6 -RATING BADGE, PETTY OFFICER, SECOND CLASS

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:53Z
Check the source to verify that this is the current version before use.

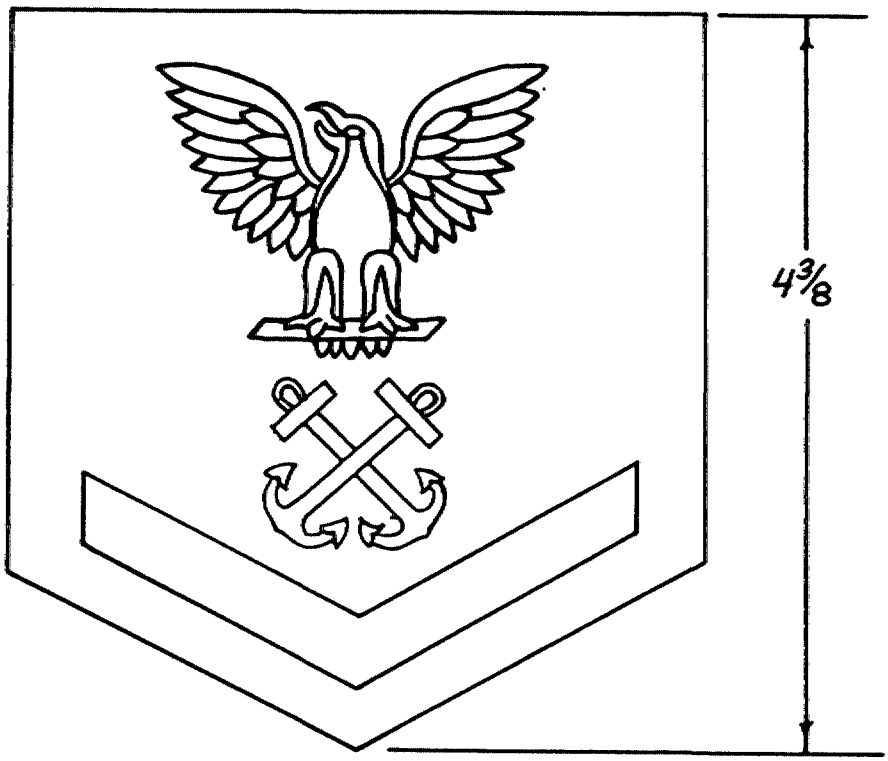
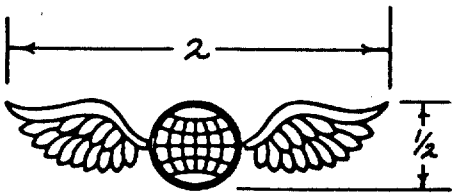
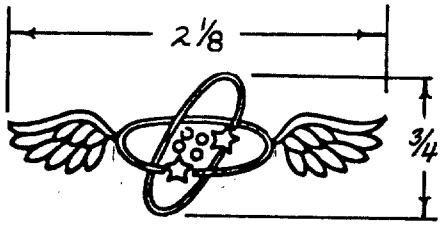


FIG. 7 - RATING BADGE, PETTY OFFICER, THIRD CLASS

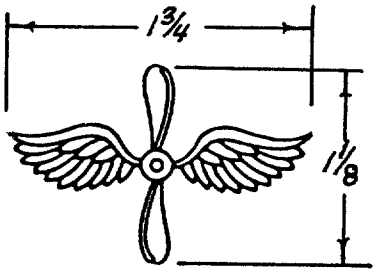
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:53Z
Check the source to verify that this is the current version before use.



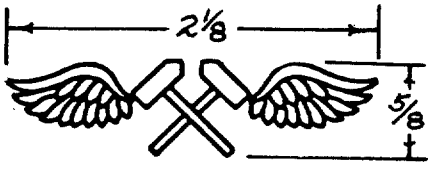
AVIATION ELECTRICIAN'S MATE (AE)



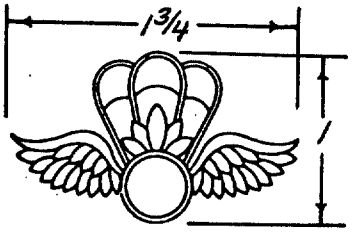
AVIATION ELECTRONIC'S TECHNICIAN (AT)



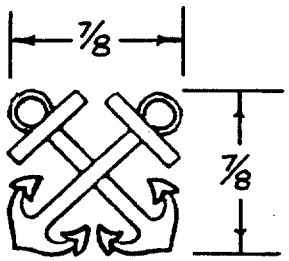
AVIATION MACHINIST'S MATE (AD)



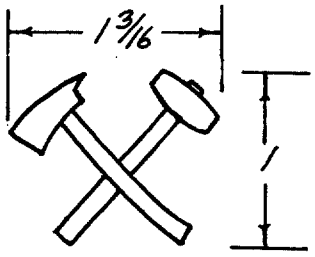
AVIATION STRUCTURAL MECHANIC (AM)



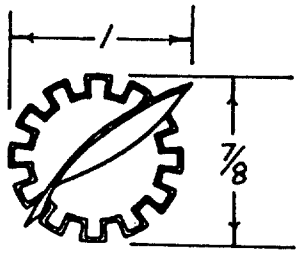
AVIATION SURVIVALMAN (ASM)



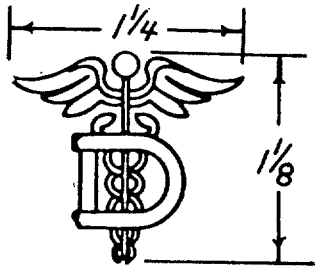
BOATSWAIN'S MATE (BM)



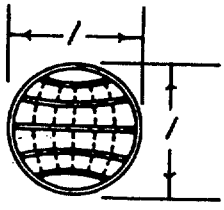
DAMAGE CONTROLMAN (DC)



DATA PROCESSING TECHNICIAN (DP)



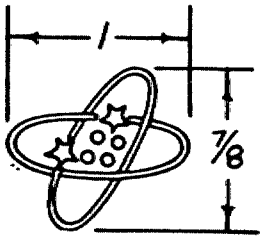
DENTAL TECHNICIAN (DT)



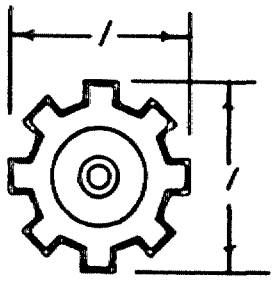
ELECTRICIAN'S MATE (EM)

FIG. 8 - SPECIALTY MARKS
STRIKER MARKS

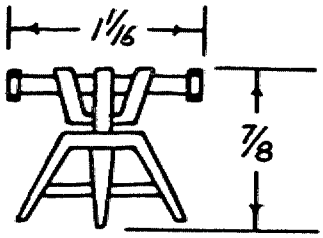
Source: https://assist.dla.mil. Downloaded: 2026-10-26T04:53Z
Check the source to verify that this is the current version before use.



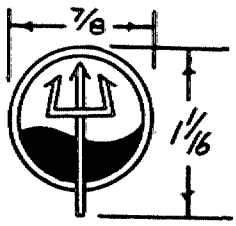
ELECTRONIC'S TECHNICIAN (ET)



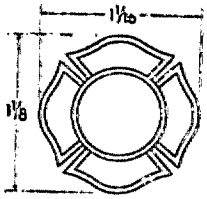
MACHINERY TECHNICIAN (MK)



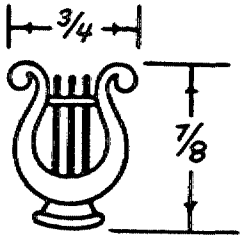
FIRE CONTROL TECHNICIAN (FT)



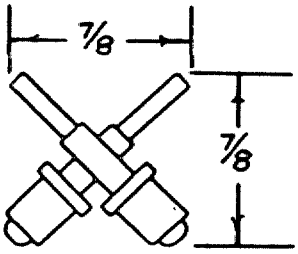
MARINE SCIENCE TECHNICIAN (MST)



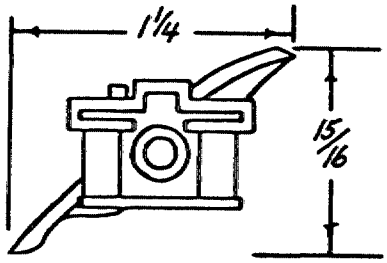
FIRE FIGHTER (FI)



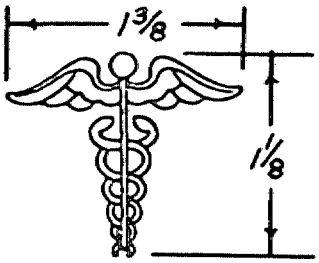
MUSICIAN (MU)



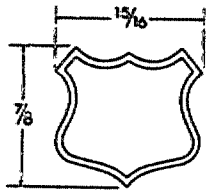
GUNNER'S MATE (GM)



PHOTOJOURNALIST (PA)



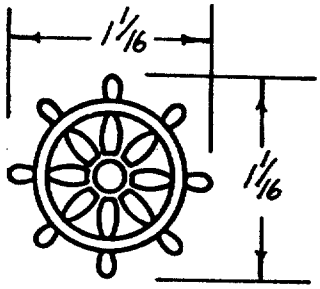
HOSPITAL CORPSMAN (HM)



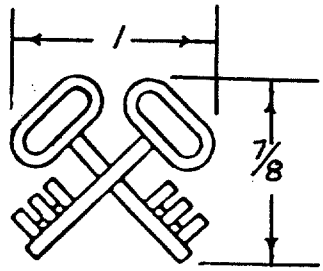
PORT SECURITYMAN (PS)

FIG. 9 - SPECIALTY MARKS
STRIKER MARKS

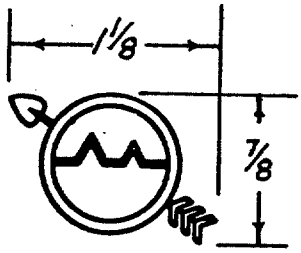
Source: https://assist.dla.mil - Downloaded: 2026-10-26T04:53Z
Check the source to verify that this is the current version before use.



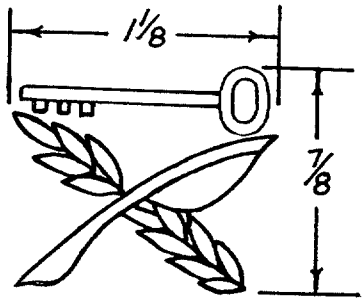
QUARTERMASTER (QM)



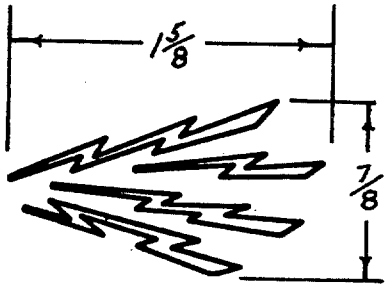
STOREKEEPER (SK)



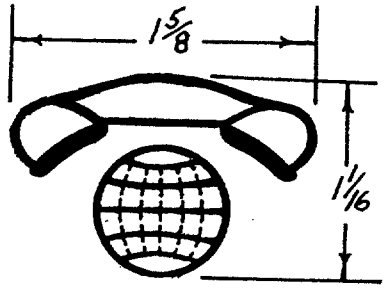
RADARMAN (RD)



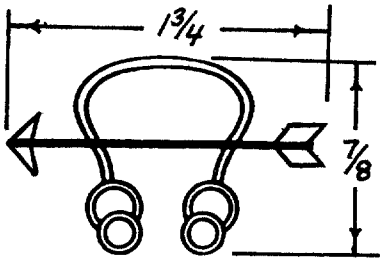
SUBSISTENCE SPECIALIST (SS)



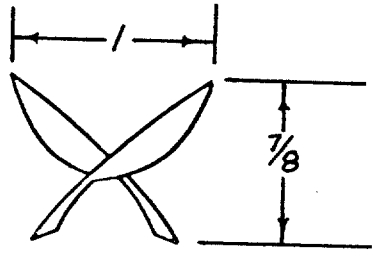
RADIOMAN (RM)



TELEPHONE TECHNICIAN (TT)



SONAR TECHNICIAN (ST)



YEOMAN (YN)

FIG. 10 - SPECIALTY MARKS
STRIKER MARKS

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:53Z
Check the source to verify that this is the current version before use.

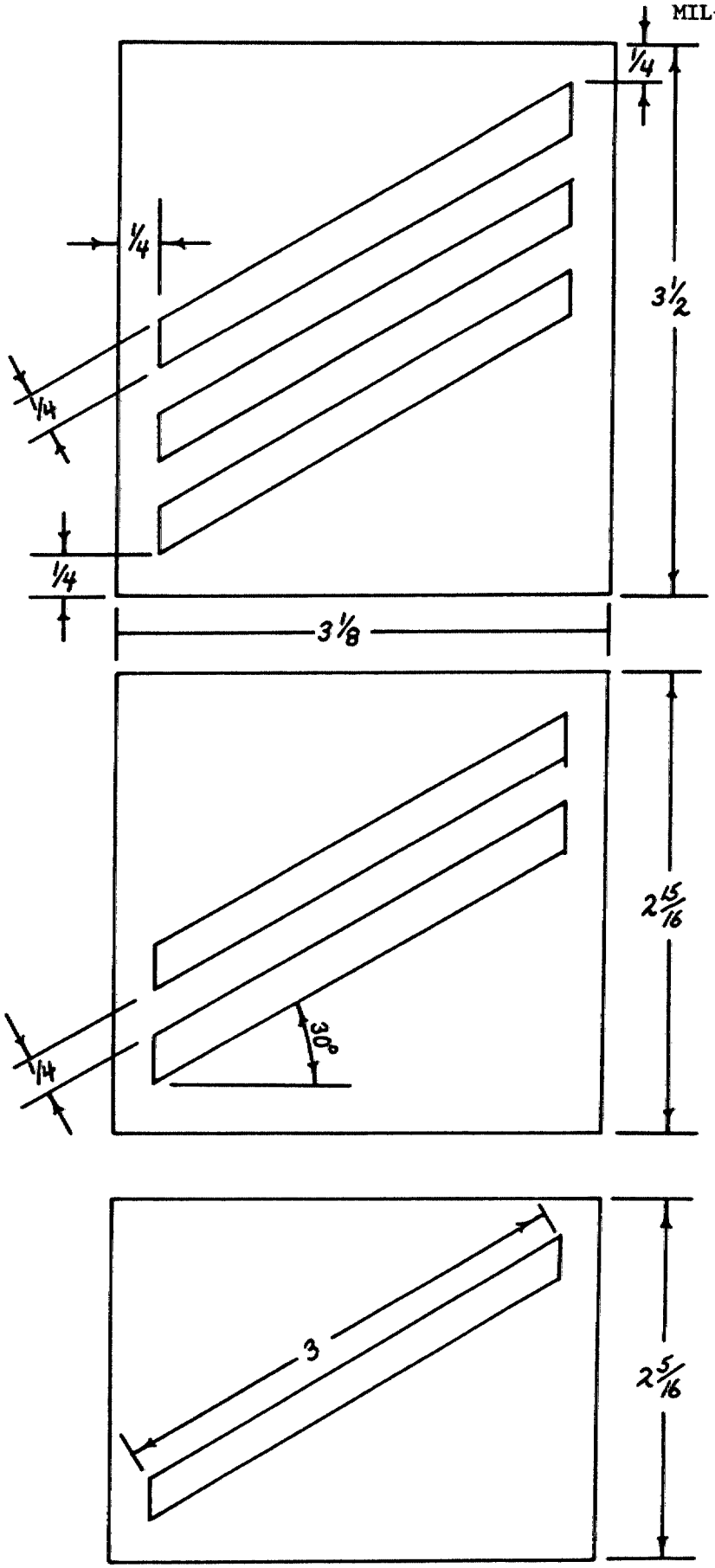


FIG. 11 - GROUP-RATE MARKS

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:53Z
Check the source to verify that this is the current version before use.

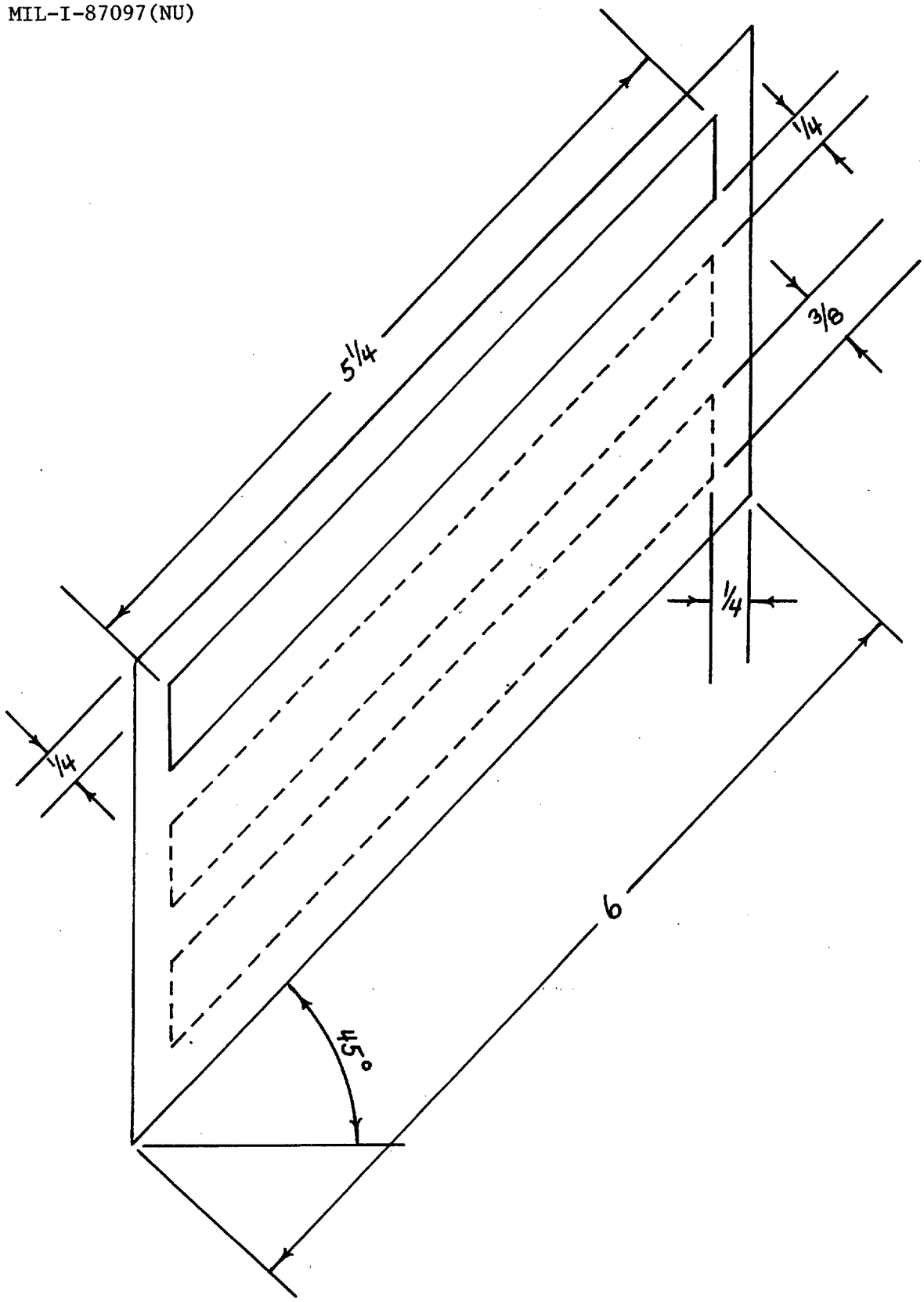
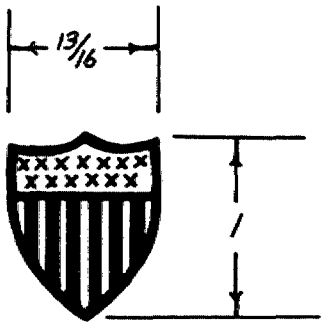
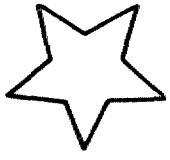


FIG. 12 - STRIPES, LENGTH OF SERVICE

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:53Z
Check the source to verify that this is the current version before use.



SHIELD
BRANCH OF SERVICE



INSIGNIA, GRADE, STAR
(3/4 INCH DIAMETER)

FIG. 13 - SHIELD AND STAR

☆ U.S. GOVERNMENT PRINTING OFFICE: 1981—703-023/5024

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:53Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-I-87097(NU) INSIGNIA, EMBROIDERED (COAST GUARD)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1426 1 OCT 76

Replaces edition of 1 Jan 72 which may be used.

S/N 0102-LF-001-4260

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:53Z
Check the source to verify that this is the current version before use.